VISUALIZING

Neural Networks

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WHAT IS A NEURAL NETWORK?

A neural network is a system of interconnected nodes (neurons) that processes data in layers — inspired by the human brain — to recognize patterns and make predictions.

Brain-Inspired

Modeled after biological neurons and synapses in the human brain.

Pattern Recognition

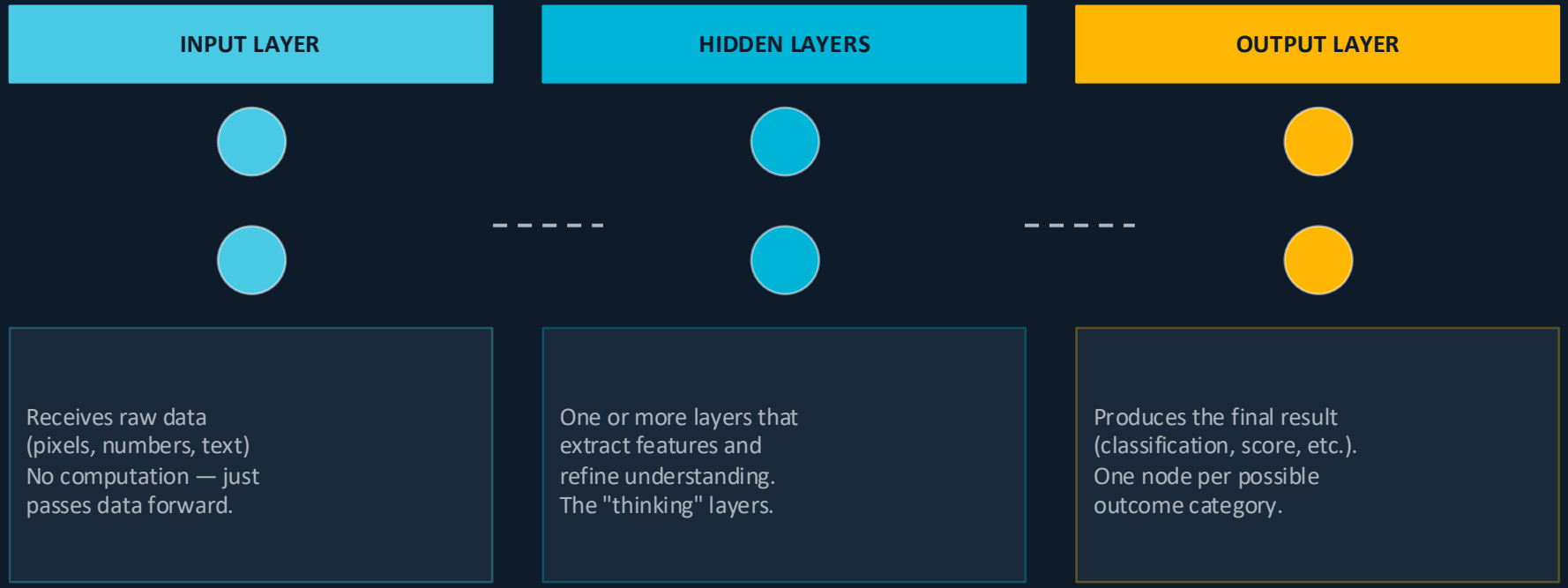
Learns to classify images, understand language, and predict outcomes.

Learns from Data

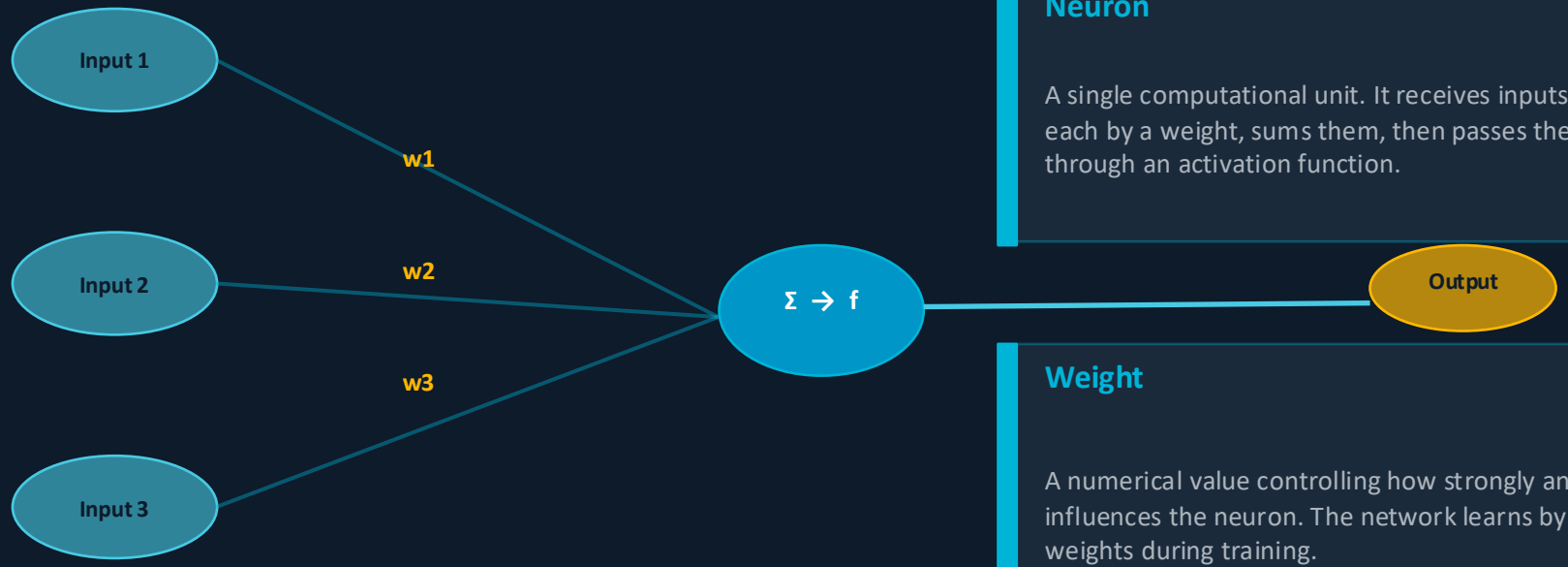
Adjusts internal parameters automatically through exposure to examples.

LAYERS

A layer is a group of neurons that work together to process and transform information as it flows through the network.



NEURONS & WEIGHTS



Neuron

A single computational unit. It receives inputs, multiplies each by a weight, sums them, then passes the result through an activation function.

Weight

A numerical value controlling how strongly an input influences the neuron. The network learns by adjusting weights during training.

ACTIVATION FUNCTIONS

Activation functions decide whether a neuron "fires" — they introduce non-linearity, allowing the network to learn complex patterns.

ReLU

$$f(x) = \max(0, x)$$

Most widely used. Fast, simple, avoids vanishing gradient. Outputs zero for negatives.

Sigmoid

$$f(x) = 1 / (1 + e^{-x})$$

Squashes output to 0–1. Useful for probability predictions in the output layer.

Tanh

$$f(x) = (e^{x} - e^{-x})/(e^{x} + e^{-x})$$

Outputs range –1 to 1. Zero-centered, often better than Sigmoid in hidden layers.

LOSS FUNCTIONS & OPTIMIZATION

📉 Loss Function

Measures how wrong the network's predictions are.
A high loss = bad predictions.
A low loss = good predictions.

Common types:
- Mean Squared Error (MSE) — for regression
- Cross-Entropy Loss — for classification

Goal: minimize the loss through training.

⚙ Optimization

Optimization algorithms update the weights to reduce loss after each batch of training data.

Most popular: Gradient Descent
- Calculates how each weight affects the loss
- Nudges weights in the direction that reduces error

Adam Optimizer — adaptive learning rate; widely used in modern deep learning.

HOW NEURAL NETWORKS LEARN

1

Forward Pass

Input data flows through every layer, producing a prediction output.

2

Calculate Loss

The loss function compares the prediction to the true answer and scores the error.

3

Backpropagation

The error signal travels backward through the network, computing weight gradients.

4

Update Weights

The optimizer adjusts each weight slightly to reduce the loss for the next pass.

COMPONENT SUMMARY

Component	Role	Example
Layer	Groups neurons; processes & passes data forward	Input, Hidden, Output layers
Neuron	Receives inputs, computes a weighted sum, outputs a value	Single node in a hidden layer
Weight	Controls influence of each input connection	$w_1 = 0.8$, $w_2 = -0.3$
Activation Function	Introduces non-linearity; determines neuron output	ReLU, Sigmoid, Tanh
Loss Function	Quantifies prediction error during training	MSE, Cross-Entropy
Optimizer	Updates weights to reduce loss over time	Gradient Descent, Adam

KEY TAKEAWAYS

Neural Networks Made Simple

- Layers group neurons and direct the flow of information.
- Neurons compute weighted sums and apply activation functions.
- Weights are the learnable parameters the network adjusts.
- Loss functions measure prediction error; optimizers reduce it.
- Training is a repeated cycle of predict → measure → adjust.